RAG Learning

Eduqas GCSE Geography B (C112QS) — Component 3: Applied Fieldwork Enquiry

4.2 Applied Fieldwork Enquiry

1. What Is Component 3?

Component 3 is a **1 hour 30 minute** written exam worth **30% of the GCSE**. There is no coursework: everything you learned on your fieldwork days is assessed through questions on this paper. In June 2024 the paper carried **76 marks**, including **4 marks for spelling, punctuation, grammar and specialist terminology (SPaG)** on the final extended question.

You must carry out **at least two fieldwork enquiries** in contrasting environments — normally one with a **physical** focus (for example a river, a beach or weather and microclimate) and one with a **human** focus (for example a town centre, regeneration, traffic or environmental quality).

Part	What it tests	June 2024 marks
Part A	An enquiry theme (in 2024, *investigating change over time*). Starts with data about an unfamiliar enquiry, then asks about **your own** fieldwork.	18
Part B	A second enquiry theme (in 2024, *investigating mitigating risk*). Same pattern: unfamiliar data first, then your own fieldwork.	18
Part C	**The wider UK dimension**: a UK-scale issue (in 2024, river flooding) using a Resource Folder. Builds to a **12-mark** evaluation plus **4 SPaG**.	36 + 4

Exam Tip — two kinds of question: Every part mixes questions about an **unfamiliar** enquiry (someone else's data, maps and methods) with questions that say *"This question is about your own experience…"*. For the second kind you **must** name your actual location, equipment, data and results. Generic answers ("we used a questionnaire and it was good") are capped in the lowest band.

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2. The Enquiry Process

The exam is built around the six stages of geographical enquiry. Questions are labelled in the mark scheme against these stages, so learning them gives you a framework for every answer.

Stage	What you do	Typical exam command
1. Enquiry question	Choose a question or hypothesis linked to geographical theory and a real place.	"Suggest one strength of your enquiry question."
2. Data collection	Collect primary and secondary data using appropriate methods and sampling, safely.	"Give one reason why your primary data collection was accurate."
3. Presentation	Process and present data using appropriate maps, graphs and diagrams.	"Which graph would be suitable...? Give one reason."
4. Analysis	Describe patterns, calculate statistics, identify anomalies, link results to theory.	"Calculate the median...", "Describe the trend..."
5. Conclusions	Answer the enquiry question using evidence.	"Suggest one conclusion... Use data in your answer."
6. Evaluation	Judge the reliability and accuracy of methods, data and conclusions; suggest improvements.	"Evaluate one strength and one weakness of your conclusions."

Writing a good enquiry question

- **Focused and testable** — "How does river velocity change downstream on the River X?" not "What is the river like?"
- **Linked to theory** — e.g. the Bradshaw model (rivers), longshore drift (coasts), Burgess land-use model or environmental quality decreasing towards the centre (urban). Theory lets you predict a result and then prove or disprove it.
- **Located precisely** — a named river, beach or town, so the results can be compared with other places.
- **Manageable** — possible to collect enough data safely in the time available.

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- Often broken into **sub-questions** so each data set answers one part.

Hypothesis: a statement that can be tested and accepted or rejected — e.g. *"Pedestrian counts decrease with distance from the town centre."* A **null hypothesis** states there is no relationship.

3. Primary and Secondary Data

	Primary data	Secondary data
Definition	Collected first-hand by you in the field	Collected by someone else and used by you
Examples	River width, depth and velocity; beach profiles; pebble size; pedestrian and traffic counts; questionnaires; environmental quality surveys; land-use maps; field sketches; photographs	Census data, Met Office climate records, Environment Agency flood maps and river levels, OS and historic maps, Google Street View, council reports, house-price websites, newspaper articles
Strengths	You know exactly how it was collected; tailored to your question; up to date	Covers long time periods and large areas; saves time; allows comparison and context
Weaknesses	Time-consuming; small samples; single visits; human error	May be out of date or collected differently; may not match your sites exactly; possible bias

Exam Tip — "Why was secondary data useful?": Credit is given for explaining or backing up primary patterns, giving context to the enquiry question, adding a **time dimension** (before and after, long-term trends), increasing validity, and covering places you could not visit. Always name the source: "Met Office records for 1991–2020 showed that...". With no secondary source identified, answers are capped at 2 marks out of 4.

Common physical methods

Method	How	Accuracy issues and improvements
River width and depth	Tape measure bank to bank; metre rule at equal intervals across the channel	Tape sagging; rule sinking into soft bed or moved by current. Keep tape taut at water level; take depth at regular, close intervals.
River velocity	Time a float over 10 m, or use a flow meter	Floats caught in eddies or wind; reaction-time error. Repeat three times and average; a flow meter is more precise.
Discharge	Cross-sectional area × velocity (cumecs)	Errors in width, depth and velocity multiply together.
Beach profile	Ranging poles and clinometer at each break of slope	Poles not vertical on soft sand; tides limit time. Use consistent intervals and the same observer.
Sediment size and shape	Measure long axis with a ruler/calipers; Powers' roundness index	Choosing pebbles by eye is biased — use random sampling within a quadrat.
Microclimate / weather	Thermometer, anemometer, rain gauge, recording sheet at fixed times	Readings change through the day; shade the thermometer; record at the same time at every site.

Common human methods

Method	How	Accuracy issues and improvements
Pedestrian / traffic count	Tally for a fixed time (e.g. 5 minutes) at each site	Varies with time of day, day of week, weather. Count at the same time at all sites; repeat on other days.
Questionnaire	Closed and open questions to the public	Small or biased samples (only approaching friendly-looking people); leading questions. Use systematic sampling (every 5th person) and pilot the questions.

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Environmental quality survey	Score factors (litter, noise, building condition) on a bi-polar scale, e.g. –3 to +3	Subjective — group members score differently. Agree clear criteria; average several scorers.
Land-use mapping	Record the use of every building on a base map	Time-consuming; categories unclear. Use a standard key (e.g. Goad categories).
Field sketch / annotated photo	Record features with labels and a title, direction and date	Selective. Add a scale, north arrow and annotations that explain, not just label.
Air quality	Lichen surveys, diffusion tubes or a handheld meter	Short sampling periods may not represent normal conditions.

4. Sampling

Technique	How it works	Strength	Weakness
Random	Sites/people chosen by random numbers	Everything has an equal chance of being chosen — no bias	May miss important areas; sites can be hard to reach
Systematic	Regular intervals — every 50 m along a transect, every 5th person	Covers the whole area evenly; good for showing change along a line or road	May miss features between intervals; can be biased if the pattern repeats at the interval
Stratified	Sample in proportion to sub-groups — e.g. each land use, age group or zone	Represents all groups fairly (e.g. different land uses and traffic)	Needs prior knowledge of the groups; more complicated
Opportunistic	Whoever or wherever is available	Quick and easy	Most likely to be biased and unrepresentative

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Exam Tip — matching questions: The 2024 paper asked candidates to match each technique to its strength: *systematic* → useful for features that change along a road or within an area; *stratified* → takes account of different land uses and traffic; *random* → everything has an equal chance of being measured. Learn these phrases.

Risk assessment

Fieldwork questions can ask about safety. Typical hazards are fast or deep water, slippery rocks, tides and cliff falls, traffic, extreme weather and talking to strangers. Reduce risk with wellies and a buddy system, checking tide times, staying out of water above knee depth, high-visibility jackets near roads, and working in groups.

5. Presenting Data

Technique	Best for	Why it is suitable
Bar chart / comparative bar chart	Discrete categories, comparing sites	Easy to compare values side by side
Line graph	Continuous data and change over time	Shows trends; never use for unrelated categories
Scatter graph + line of best fit	Relationship between two variables	Shows correlation and anomalies
Dispersion graph	Spread of values at different sites	Shows range, median, clustering and extremes; compares data sets
River cross-section	Channel shape	Visual representation of the river bed (plot depth downwards)
Kite diagram	Vegetation cover along a transect	Shows how cover changes with distance
Radial / rose diagram (wind rose)	Direction-based data — wind, orientation	Shows the direction with most occurrences and greatest strength

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Flow-line map	Movements of people or traffic	Arrow width proportional to flow, located on a map
Proportional symbols / located bar charts	Values at map locations	Shows the spatial pattern and the size of values together
Choropleth map	Rates by area (e.g. census data)	Shows spatial patterns; hides variation within each area
Pie chart / divided bar	Parts of a whole (percentages)	Shows proportions; poor with many categories

Good presentation includes a **title**, labelled axes with **units**, a **key**, an appropriate scale, and on maps a **north arrow** and **scale**. Exam questions often ask you to suggest improvements to a figure — these features are the usual answers, along with making symbols proportional and giving exact dates and values instead of "up to" or "now/later".

6. Analysis — the Maths You Need

Statistic	How to calculate	Example
Mean	Add all values ÷ number of values	28, 26, 22 → 76 ÷ 3 = 25.3
Median	Put in order; middle value (average the middle two if an even number)	−21, −16, −16, +200 → (−16 + −16) ÷ 2 = −16
Mode	Most frequent value	—
Range	Highest − lowest	8.5 − 3.4 = 5.1
Interquartile range	Upper quartile − lower quartile (median of top half − median of bottom half)	Shows spread without extremes
Percentage change	(new − old) ÷ old × 100	13 °C → 15 °C: 2 ÷ 13 × 100 = 15.4%

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Ratio / proportion	Simplify by dividing both sides	1.8 : 7.2 = 1 : 4
Speed	Distance ÷ time	10 m ÷ 25.3 s = 0.4 m/s

Exam Tip — median versus mean: When data contains an extreme outlier (e.g. +200 among values around −16), the **median** is more appropriate because one extreme value skews the mean. Always show your working — there is usually 1 mark for the method and 1 for the answer, and negative signs and units matter.

Reading maps

- **Six-figure grid references:** eastings (along the corridor) first, then northings (up the stairs), estimating tenths of each square — e.g. 097213.
- **Area:** count grid squares — on a 1:25 000 or 1:50 000 OS map each square is 1 km², so a study area inside a town ring road is likely to be about 1 km², not 10 m² or 10 km².
- **Describing patterns:** give the general trend, then use data and name places; mention exceptions (anomalies).

7. Conclusions and Evaluation

Drawing conclusions

1. Answer the enquiry question directly — was the hypothesis supported, partly supported or rejected?
2. Support it with **specific data**: "velocity increased from 0.12 m/s at site 1 to 0.41 m/s at site 6".
3. Explain the result using **geographical theory**.
4. Note **anomalies** and suggest reasons for them.
5. State the significance — what the result means for the place, for theory, or for people.

Evaluating — accuracy, reliability and validity

Accuracy: how close a measurement is to the true value (affected by equipment and human error).

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Reliability: whether repeating the method would give the same results (affected by sample size, repeats and timing).
Validity: whether the data actually answers the enquiry question.

Weakness	Effect	Improvement
Single visit on one day	Results may reflect unusual conditions (weather, flow, day of week)	Repeat at different times / seasons and average
Small sample	Unrepresentative; anomalies have a large effect	More sites, more questionnaire responses, more repeats
Human error	Inaccurate measurements (reaction times, misread instruments)	Use digital equipment (flow meter); the same person takes each reading
Subjective scoring	Results vary between observers	Clear criteria; average several observers
Biased sampling	Some groups over- or under-represented	Random, systematic or stratified sampling
Sites too few or poorly spaced	Pattern unclear	Systematic intervals along the whole transect

Exam Tip — "Evaluate one strength and one weakness of your conclusions" (4 marks): Make a point and develop it twice for each side. Example: *"A strength was that my conclusion directly answered my question, showing that velocity increased downstream as the Bradshaw model predicts, and was supported by data from all six sites. A weakness was that we only visited once in dry weather, so discharge was low and the pattern might be different in winter, which makes the conclusion less reliable."*

8. Your Own Fieldwork — Prepare These Before the Exam

Write a one-page summary of each enquiry and learn it. Questions on your own fieldwork can appear in Parts A and B whichever theme is set, so you need the facts ready to adapt.

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Heading	What to learn (for each enquiry)
Location	Exact place name, why it was chosen (safe, accessible, shows the processes in the theory)
Enquiry question / hypothesis	Wording; the theory it is linked to; one strength of the question
Primary methods	Equipment, sampling technique, number of sites, how you made it accurate
Secondary data	At least one named source and exactly how it helped
Presentation	One technique you used and why it was suitable; one improvement
Results	Two or three actual figures you can quote
Conclusion	The answer to your question, with evidence
Evaluation	Two strengths and two weaknesses, each with an improvement

Worked example — physical enquiry (rivers): "How do channel characteristics change downstream on a small river?" Six sites chosen systematically every 500 m from source; width (tape), depth (metre rule every 20 cm), velocity (float timed over 10 m, three repeats); Environment Agency river-level data used to check the river was at normal flow. Presented as cross-sections and scatter graphs of distance against discharge. Conclusion: width, depth and discharge increased downstream, supporting the Bradshaw model, with an anomaly at a riffle where the channel was shallow. Evaluation: floats were caught in eddies, so a flow meter would be more accurate; one visit in summer limits reliability.

Worked example — human enquiry (urban): "Does environmental quality improve with distance from the town centre?" Ten sites along a transect, systematic every 200 m; environmental quality survey (−3 to +3 on eight factors), pedestrian counts for five minutes, 30 questionnaires; census deprivation data as secondary data. Presented as a located bar chart on a map and a scatter graph. Conclusion: quality generally improved outwards, but a site near a main road scored low

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because of traffic noise. Evaluation: scoring was subjective, so averaging several scorers and visiting at different times would improve reliability.

9. Part C — The Wider UK Dimension

Part C uses a Resource Folder to explore a geographical issue at the scale of the whole UK. In June 2024 the issue was **river flooding**: a graph of UK rainfall since 1836, UK population projections to 2045, a flood-risk map of Marlow on the River Thames and a diagram of land-use zoning on a floodplain. The questions moved from reading graphs and maps to explaining processes, and finished with:

> *"Land use zoning is the best way to sustainably manage river flood risk in the UK. To what extent do you agree with this statement?"* [12 + 4 SPaG]

Part C draws on your knowledge from Component 1 (rivers, coasts, weather, climate change, UK cities and population), so revise those themes at UK scale — not just your case studies.

Answering the 12-mark evaluation

1. **State your position** in the opening sentence (agree, partly agree, disagree).
2. **Argument** — two or three developed points for the statement, each using evidence from the Resource Folder (quote figures and place names).
3. **Counter-argument** — limitations and alternatives, again with evidence.
4. **Sustainability** — explicitly consider social, economic and environmental effects, short and long term. The top band asks for this.
5. **Wider knowledge** — bring in examples from your studies (e.g. Pickering "Slowing the Flow", the Jubilee River, Storm Desmond).
6. **Conclusion** — a justified judgement that follows from your argument; "it depends" must say *on what*.

Exam Tip — SPaG: The 4 SPaG marks reward accurate spelling, punctuation and grammar and a **wide range of specialist terms**. Use words such as *impermeable surfaces, lag time, peak discharge, floodplain zoning, hard and soft engineering, sustainability, stakeholders* accurately.

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